16.07 Dynamics — Quiz 4: Kane Damper

Name: _____

The questions below refer to the rigid body with a Kane damper from Problem Set 4. Recall that the fluid applies a torque $C_d \left({}^B\boldsymbol{\omega}_d - {}^B\boldsymbol{\omega}\right)$ to the body, and that the total kinetic energy changes at the rate

$$\dot{T} = -C_d \left\| {}^B\boldsymbol{\omega}_d - {}^B\boldsymbol{\omega} \right\|^2 .$$

1. Without a damper, spin about which principal axes is stable? Which of these does the Kane damper make unstable, and why?

2. What happens to the rate of energy dissipation in the limit $C_d \to 0$? What about the limit $C_d \to \infty$?

3. A body with principal inertia matrix $\mathbf{J} = \mathrm{diag}(2,\ 3,\ 4)$ contains a Kane damper with $J_d = 1$. It starts in an equilibrium spin at rate Ω about $\hat{b}_1$ (the minor axis), with the damper co-rotating, and is then given a tiny disturbance. About which axis, and at what angular velocity, does the body eventually spin?